FIGURE · ALGORITHM 3

Homomorphic Functional Inner Product Score Computation

A geometric view of server-side encrypted scoring — the FIP Verifier combines encrypted encoded client coordinates with the encoded trusted reference, decrypts only the scalar accumulator through the TKA, decodes the signed fixed-point scalar score, and returns the round score map $\mathcal{S}^{(t)}$.

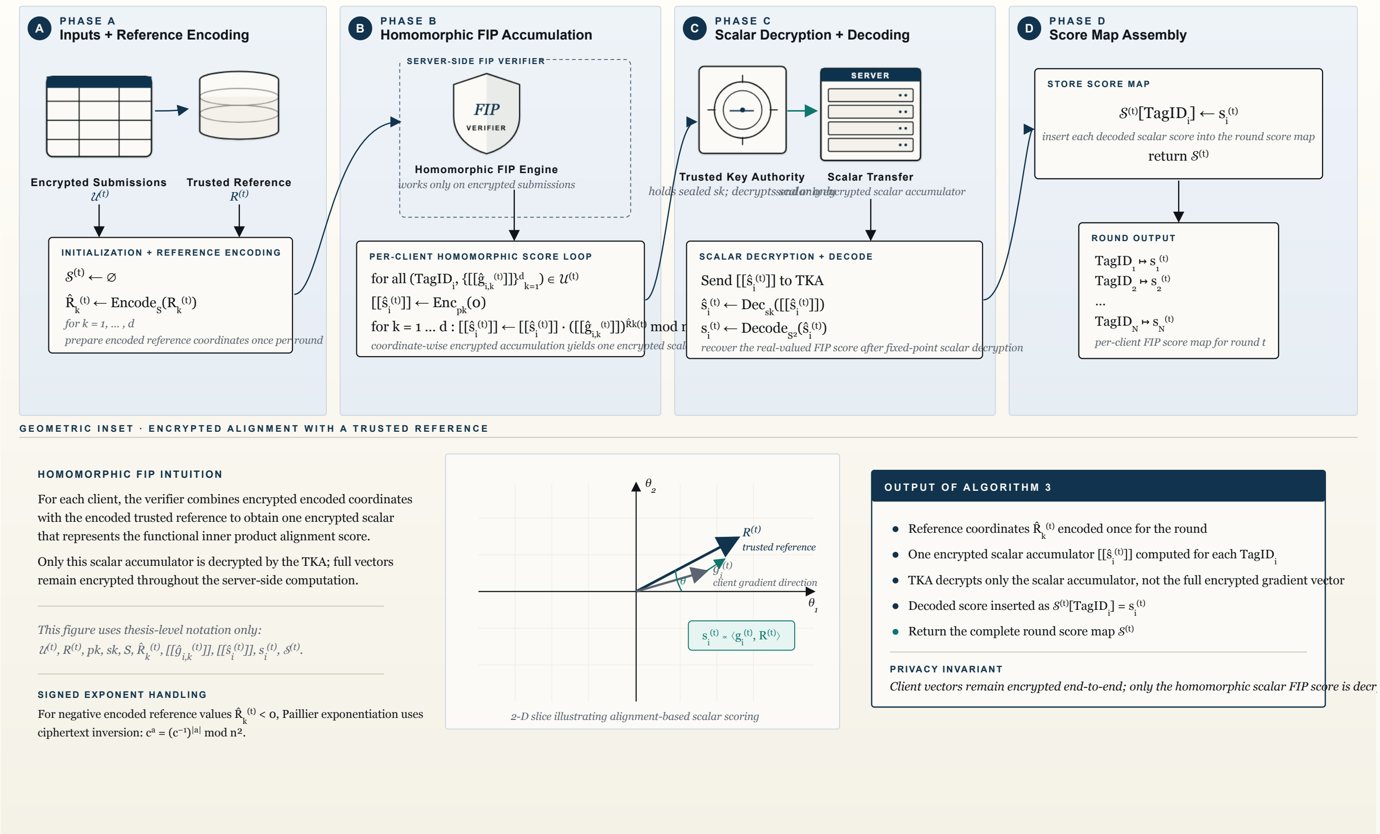

Figure 3. Algorithm 3. The FIP Verifier receives the encrypted submission set $\mathcal{U}^{(t)}$ and trusted reference $R^{(t)}$. It initializes the score map $\mathcal{S}^{(t)} = \varnothing$, encodes the reference coordinates as $\hat{R}_k^{(t)}$, and for each client accumulates an encrypted scalar inner-product score $[[\hat{s}_i^{(t)}]]$ using Paillier homomorphism. Only this scalar accumulator is sent to the TKA for decryption; the TKA returns $\hat{s}_i^{(t)}$, which is decoded as $s_i^{(t)}$ using $\mathrm{Decode}_{S^2}(\cdot)$. The verifier stores the result as $\mathcal{S}^{(t)}[TagID_i] \leftarrow s_i^{(t)}$. The inset illustrates alignment-based scalar scoring with respect to the trusted reference direction.